MIL-P-12764G (AR)
11 March 1981
SUPERSEDING
MIL-P-12764F (MU)
10 May 1968

MILITARY SPECIFICATION

PROJECTILE, 4.2 INCH, ILLUMINATING, M335A2 LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not covered by the drawings and provides quality assurance provisions for the loading, assembling and packing of one type of projectile designated as Projectile, 4.2 Inch, Illuminating, M335A2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

FEDERAL

FED-STD-151 - Federal Test Method Standard

^U.S. GOVERNMENT PRINTING OFFICE 1981—703-023/1971

FSC: 1315

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-STD-105 -Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-286 - Propellants, Solids: Sampling Examination and Testing

MIL-STD-1234 - Pyrotechnics: Sampling, Inspection and Testing

DRAWINGS (See 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

PRODUCT AND PACKAGING DRAWINGS

8848553 - Packaging and Marking of Pellet for 4.2 Inch Mortar Shell Bodies

8861520 - Marking Diagram for Pallets for Projectile 75MM to 120MM

8886446 - Illuminating Canister Loading Assembly

8886453 - Projectile, for Cartridge, 4.2 Inch, Illuminating, M335A2 Loading Assembly

INSPECTION EQUIPMENT DRAWINGS

9201136 - Tunnel, Light

9201268 - Procedure, Light Output Measurement

9201390 - Procedure, Photocell Checkout

9200964 - Photometer Head with Cell

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on drawing (dwg.) 8886453, and associated drawings and with all requirements specified in applicable specifications and standards.

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3.3 Moisture content

3.3.1 First fire composition. The moisture content of the first fire composition at the loading station, prior to loading, shall not exceed 0.30 percent.

3.3.2 Illuminating composition. The moisture content of the illuminant composition at the loading station, prior to loading, shall not exceed 0.30 percent.

3.4 Static test of illuminating canister loading assembly

3.4.1 Burning time. The burning time shall not be less then 65 seconds (sec).

3.4.2 Luminosity. The average luminosity shall not be less then 703,000 candlepower and no assembly shall have a luminanosity of less than 670,000 candlepower.

3.5 Proving ground.

3.5.1 Functioning. The projectile loading assembly shall function satisfactorily (see 6.5) and shall provide effective illumination (see 6.6) for 60 sec. minimum (min.).

3.5.2 Premature bursts and security of parts. There shall be no premature burst or evidence of metal parts failure in the bore of the weapon or in approximately the first 200 yards horizontal range

3.6 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specifed in the contract.

3.7 Workmanship. The parts and assemblies shall be free of burrs, chips, sharp edges, cracks, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material.

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4. QUALITY ASSURANCE PROVISION

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specifed herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on these items:

a. First article inspection
b. Quality conformance inspection

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. This requirement is applicable when the loading of projectile and assembly into cartridges is accomplished at different contractor factilities. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Pin, shear	8833744	15
Support, lower	8886452	15
Plate, baffle	9234272	15
Canister body	8886449	15
Canister body assembly	8885176	5
Illuminating canister loading assembly	8886446	15
Bag, parachute	8885169	15
Parachute assembly	8885180	15
Parachute, shroud protector and bag assembly	8885182	15
Projectile loading assembly	8886453	15
First fire composition	8886446	4 ounces (ozs.)
Illuminating composition	8886446	4 ounces (ozs.)

4.3.2 Inspection to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-12764G (AR)

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
	Projectile Loading Asembly and Components		1 OF 3	See Below
				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH / PARAGRAPH REFERENCE /INSPECTION METHOD
	Pin, shear (dwg. 8833744)			
	Examination for defects	15		3.2 4.4.2.1
	Support, lower (dwg. 8886452)			
	Examination for defects	15		3.2 4.4.2.2
	Plate, baffle (dwg. 9234272)			
	Examination for defects	15		3.2 4.4.2.3
	Body, canister (dwg. 8886449)			
	Examination for defects	15		3.2 4.4.2.4
	Hardness test	5		3.2 4.5.3
	Illuminating canister loading assembly prior to load of illuminent (dwg. 8886446)			
	Examination for defects	15		3.2 4.4.2.5
	Salt spray test	2		3.2 4.5.5

NOTES:

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Projectile Loading Asembly and Components	2 OF 3	NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Illuminating canister loading assembly (dwg. 8886446)				
	Examination for defects	15		3.2	4.4.2.6
	Static test	3		3.4	4.5.2
	Moisture content of first fire composition	4 ozs		3.3.1	4.5.1
	Moisture content of illuminant composition	4 ozs		3.3.2	4.5.1
	Parachute assembly (dwg. 8885180)				
	Examination for defects	15		3.2	4.4.2.7
	Bag, parachute (dwg. 8885169)				
	Examination for defects	15		3.2	4.4.2.8
	Parachute, shroud protector and bag assembly (dwg. 8885182)				
	Examination for defects	15		3.2	4.4.2.9

NOTES:

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-12764G (AR)

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER
	Projectile Loading Asembly and Components				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Projectile, loading assembly (dwg. 8886453)				
	Examination for defects	15		3.2	4.4.2.12
	Proving ground test	15		3.6	4.5.4

NOTES:

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of loaded projectile assemblies shall contain:

a. Projectile metal parts assemblies of one lot interfix from one maufacturer;

b. Parachute assemblies of one lot interfix from one manufacturer;

c. Illuminating canister loading assemblies of one lot from one manufacturer.

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-12764G (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8833744
4.4.2.1	Pin, Shear				NEXT HIGHER ASSEMBLY 8886453
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter, min.		0.40%	3.2	Gage
102	Length, min.		0.40%	3.2	Gage
Minor	None defined				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-12764G (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8886452
4.4.2.2	Support, Lower				NEXT HIGHER ASSEMBLY 8886453
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length		0.40%	3.2	Gage
102	Radius		0.40%	3.2	Gage
103	Arc length		0.40%	3.2	Gage
104	Presence of sharp corners or edges		0.40%	3.2	Visual
Minor	None defined				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-12764G (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Plate, Baffle	1 OF 1	9234272
			NEXT HIGHER ASSEMBLY: 8886453

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of plate		0.40%	3.2	Gage
102	Diameter of hole, min.		0.40%	3.2	Gage
103	Location of holes		0.40%	3.2	Gage
Minor	None defined				

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-12764G (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Body, canister	1 OF 1	8886449
			NEXT HIGHER ASSEMBLY: 8886453

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Wall thickness, min.		0.40%	3.2	Gage
102	Outside diameter, min.		0.40%	3.2	Gage
103	Hole diameter, max.		0.40%	3.2	Gage
104	Length		0.40%	3.2	Gage
105	Hardness	*		3.2	4.5.3
Minor	None defined				

NOTES: *For continuous heat treat process, samples shall be selected from each eight hours of production in accordance with MIL-STD-105, Table IIA, code letter D and an AQL of 1.5%. If a batch heat treat process is used the samples shall be selected as specified above from each batch.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-12764G (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Illuminating Canister Loading Assembly Prior to Loading of Illuminant				8886446
					NEXT HIGHER ASSEMBLY 8886453
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Brake missing		0.40%	3.2	Visual
102	Spot or projection weld missing		0.40%	3.2	Visual
103	Brake above flush		0.40%	3.2	Visual
104	Laminac present on open end of can between brakes		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-12764G (AR)

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.6	Illuminating Canister Loading Assembly		1 OF 1	8886446
				NEXT HIGHER ASSEMBLY
				8886453

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Depth to first fire surface, max.		0.40%	3.2	Gage
102	Swivel pin fails to rotate freely		0.40%	3.2	Manual
103	Cover disc loose, missing or improperly assembled		0.40%	3.2	Visual/Manual
104	Disc not flush to one disc above flush		0.40%	3.2	Visual
105	First fire composition moisture content	*		3.3.1	4.5.1.1
106	Iluminant composition moisture content	*		3.3.2	4.5.1.2
107	Static test	3		3.4	4.4.3.1/4.5.2
Minor					
201	Outside diameter, max.		0.65%	3.2	Gage
202	Marking improper		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES: A four ounce sample from each eight hours of production shall be selected. If the sample after analyses fails to comply with the applicable requirement the eight hours of production represented by the sample shall be rejected.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-12764G (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Parachute Assembly	1 OF 1	8885180
			NEXT HIGHER ASSEMBLY 8886453

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length of shroud line to reinforcing tape stitches		0.40%	3.2	Gage
102	Stitch type incorrect		0.40%	3.2	Visual
103	Stitch missing or broken		0.40%	3.2	Visual
104	Holes in canopy segment greater than 1/8 inch in diameter		0.40%	3.2	Visual
105	Shroud cord twisted		0.40%	3.2	Visual
Minor					
201	Length of sleeving shroud loop		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-12764G (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.8	Bag, Parachute		8885169
			NEXT HIGHER ASSEMBLY 8886453

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length from closed end of bag to start of closure stitching		0.40%	3.2	Gage
102	Length of closure stitching to right hand edge of top flap		0.40%	3.2	Gage
103	Length of overlap at open end		0.40%	3.2	Gage
104	Width of bag (at development)		0.40%	3.2	Gage
105	Length of bag (at development)		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-12764G (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8885182
4.4.2.9	Parachute, Shroud Protector and Bag Assembly				NEXT HIGHER ASSEMBLY 8886453
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Attachment loop improperly located from lapped edges of shroud line protector (Step IX)		0.40%	3.2	Visual
102	Improper number of shroud cord loops (Step VIII)		0.40%	3.2	Visual
103	Improper number of rubber bands per loop (Step VIII)		0.40%	3.2	Visual
104	Improper number of accordian folds (Step III)		0.40%	3.2	Visual
105	Improper number of folds (Step I)		0.40%	3.2	Visual
106	Gore not flush with hemline (Step I)		0.40%	3.2	Visual
107	Split not opposite shroud lines (Step X)		0.40%	3.2	Visual
Minor					
201	Width of pack, max. (Step X)		0.65%	3.2	Gage
202	Length of pack, max. (Step X)		0.65%	3.2	Gage

NOTES: The inspections are to be performed on completed lots of parachute and shroud protector bag assemblies. After inspection the parachutes may be refolded, inserted into the bags, and returned to the lot.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-12764G (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Projectile Loading Assembly Prior to Assembly of Canister and Parachute Assembly	1 OF 1	8336453
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Baffle plate missing		0.40%	3.2	Visual
Minor 201	Foreign material in shell prior to assembling baffle plate		0.65%	3.2	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-12764G (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Parachute and Illuminating Canister Assembly Prior to Loading into Projectile	1 OF 1	8886453
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Lower support missing		0.40%	3.2	Visual
102	Rubber band not removed from shroud line protector		0.40%	3.2	Visual
103	Parachute wrapper flaps improperly located with respect to attachment loop		0.40%	3.2	Visual
Minor					
201	Parachute wrapper torn or missing		0.65%	3.2	Visual
202	Canister wrapper torn or missing		0.65%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-12764G (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.12	Projectile Loading Assembly				8886453
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Shear pin missing		100%	3.2	Visual
Major					
101	Shear pin not staked		0.40%	3.2	Visual
102	Twist pin not staked		0.40%	3.2	Visual
103	Pin above flush with outer surface		0.40%	3.2	Visual
104	Gap between projectile and base, max.		0.40%	3.2	Gage 4.4.3.2/4.5.4
105	Proving ground test	48/15			
Minor					
201	Marking missing, misleading or unidentifiable		0.65%	3.2	Visual

NOTES:

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MIL-P-12764G (AR)

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.4.2.13	Pallet, Prior to Strapping		1 OF 1		8848553
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Liner missing		0.65%	5.1	Visual
202	Sleeve missing		0.65%	5.1	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-12764G (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.14	Pallet				8848553
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Strapping loose			5.1	Manual
102	Boards broken or missing			5.1	Visual
Minor					
201	Marking missing, misleading or unidentifiable			5.1	Visual

NOTES:

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4.4.3 Testing.

4.4.3.1 Static test of illuminant canister assembly. (see 3.4). The following sampling plan will apply with the sample tested as specified in 4.5.2.

4.4.3.1.1 Each inspection lot will consist of eighty five candles produced consecutively by the mixing pressing operation and will be designated as one mix/batch (see 6.7) for identification purposes.

4.4.3.1.1.1 Select a sample of three candles from each inspection lot (85 candles). The canister assembly shall be observed for compliance with the requirements specified in 3.4. Continue this sampling frequency until 15 consecutive samples (15 groups of 3 samples) comply with the specified requirements.

4.4.3.1.1.2 After successful completion of 4.4.3.1.1.1, reduce the sampling frequency to three candles from each group of six inspection lots (510 candles). The canister assembly shall be observed for compliance with the requirements specified in 3.4. Continue this sampling frequency until 15 consecutive samples (15 groups of 3 candles) comply with the specified requirements. In the event that three consecutive lots fail to comply with the requirements the sampling shall revert to 4.4.3.1.1.1.

4.4.3.1.1.3 After successful completion of 4.4.3.1.1.2, reduce the sampling frequency to three candles from each group of fifteen inspection lots (1275 candles). The canister assembly shall be observed for compliance with the requirements specified in 3.4. Continue this sampling frequency until 15 consecutive samples (15 groups of 3 candles) comply with the specified requirements. In the event that three consecutive lots fail to comply with the requirements the sampling shall revert to 4.4.3.1.1.2.

4.4.3.1.1.4 After successful completion of 4.4.3.1.1.3 reduce the sampling frequency to three candles from each eight hours of production or 1615 candles, whichever is less. The canister assembly shall be observed for compliance with the requirements specified in 3.4. In the event that three consecutive lots fail to comply with the requirements the sampling shall revert to 4.4.3.1.1.3.

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4.4.3.1.1.5 In the event of any production stoppage greater than one week, the sampling plan shall revert to 4.4.3.1.1.2 and in the event of production stoppage greater than two weeks, the sampling plan shall revert to 4.4.3.1.1.1. If the continuous mixing operation has not been shut down, but merely diverted to another item, the sampling plan shall revert to 4.4.3.1.1.2.

4.4.3.1.1.6 In the event of failure of any sample to comply with the requirements, each individual inspection lot within the mix/batch must be tested. Failure of an inspection lot to comply with the requirements shall result in rejection of that particular inspection lot.

4.4.3.2 Proving ground.

4.4.3.2.1 First Article. The fifteen sample projectiles shall be assembled into cartridges and be fired in three groups of five rounds each in accordance with 4.5.4. Observations shall be made for compliance with the criteria specified in Table I. In addition, the proving ground shall make the necessary arrangements to recover malfunctioning rounds or parachute and illuminant assemblies which may occur during the test. Arrangements for the recovery of malfuctions shall be for the first article samples only unless otherwise directed by the procuring activity.

4.4.3.2.2 Production

4.4.3.2.2.1 Beginning with the first lot produced and continuing until three consecutive lots have been found to be acceptable, forty eight sample projectiles shall be randomly selected from each lot for this test. The projectiles shall be assembled into cartridges and be fired in three groups of sixteen rounds each in accordance with 4.5.4. The projectiles shall be observed for compliance with the criteria specified in Table I.

4.4.3.2.2.2 After three consecutive lots have complied with the criteria of 4.4.3.2.2.1, fifteen sample projectiles shall be randomly selected from each lot for this test. The projectiles shall be assembled into cartridges and be fired in three groups of five rounds each in accordance with 4.5.4. The projectiles shall be observed for compliance with the criteria specified in Table I. In the event that three consecutive lots fail to comply with the requirements the sampling shall revert to 4.4.3.2.2.1.

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TABLE I

Category	Defect Sample Size	Acceptance Criteria 48 accept	48 reject	15 accept	15 reject
Critical	Premature burst and parts security	0	1	0	1
Major	Effective illumination for 40 seconds or less		(see a)		(see b)
Minor	Effective illumination for more than 40 seconds but less than 60 seconds		(see a)		(see b)

(a) The lot shall be rejected if the total number of major and minor defects is 7 or more or the number of major defects in 3 or more.

(b) The lot shall be rejected if the total number of major and minor defects is 5 or more or the number of major defects is 2 or more.

NOTE: Any projectile that fails to comply with the requirement for effective illumination due to low burst height shall be disregarded and a substitute sample shall be fired.

A premature burst attributable to premature fuze functioning or an expelling charge failure shall not be cause for rejection or assessed against the projectile

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection

4.5.1 Moisture content

4.5.1.1 First fire composition. The moisture content shall be determined in accordance with MIL-STD-1234, Method

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101.3. The special solvent to be used shall be equal parts by volume of dimethylformide and anhydrous methanol.

4.5.1.2 Illuminant composition. The moisture content shall be determined in accordance with the Karl Fischer Method as specified in MIL-STD-1234, Method 101.2 up to 5.3. A 50 ml sample of 1:1 dimethylformamidemethyl alcohol solution shall be added into a 250 ml. Erlenmeyer flask (fitted with rubber stopper) containing 10 grams of the illuminant sample. (Exercise care to exclude atmospheric moisture). The flask shall be stoppered and shaken vigorously to thoroughly mix solution and illuminant sample. Allow the solution to set 5 to 15 minutes. A blank sample of the 1:1 dimethylformamide-methanol solution shall be prepared as described above. A 60 ml. portion of methanol shall be transfered through the sample porthole into the dry tetration beaker containing a plastic coated magnet. Stopper at once. The magnet stirrer shall be started and the water back titrated in this solvent by adding a slight excess of Karl Fischer reagent to produce a reddish-brown color which persists for at least a half minute. The solution shall be titrated slowly with the water-in-methonal solution until the exact preliminary end point is reached. All succeeding operations shall be completed with 3.5-0.25 minutes. 25 ml. of the supernatant solution shall be rapidly introduced, accurately pipetted from the supernatant liquid into the beaker through the same porthole of the rubber stopper. The porthole shall be closed immediately and the contents of the beaker stirred for thirty sec. by means of the magnetic stirrer. A slight excess (10-15 ml) of the Karl Fischer reagent shall be added and carefully back-titrated with the standard water-in-methanol solution to the final and exact end point. A water determination of the dimethylformamide-methanol blank solution shall be conducted as specified above for the extraction of water from the sample. The percent water in the pyrotechnic sample shall be calculated as follows:

Calcultion: $\frac{AB}{5(10)}$ Percent H_2O

Less Std Blank

Leaves % H_2O in Sample

where:
A = mls of Karel Fischer to titrate sample
B = standardization factor (mg./ml.)
10= constant factor
5 = standard sample weight

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4.5.1.3 Alternate method for moisture content of first fire and illuminant composition. The moisture content shall be determined in accordance with Method T101.4 of MIL-STD-286. The temperature setting shall be 100°C. and the time setting 30 min.; the sample size shall be 50 plus or minus 10 grams.

4.5.2 Static test of illuminant canister assembly. The equipment required for this test is delineated in 2.1 under Inspection Equipment Drawings. A sample of three candles shall be selected to represent the mix/batch of candles (see 6.7). In calculating the candlepower of any one illuminant, the best continuous sixty seconds of illuminant readings are to be used. The first five disc assemblies should be removed prior to testing. The distance from the photometric cell to the the flare shall be 80 ± .25 feet. The axial center of the fireball shall be located 54 ± .5 inch from the floor. The temporary back wall listed on dwg. 9201136 shall be removed so that the distance from the center of the candle to the back wall is from 14 to 15 feet. The center line of the baffle opening shall be in line with the axial center of the fire ball and the axial center of the photometric cell. The test shall be conducted in a hearth that has been painted flat black. Depending on the ash build-up, the tunnel hearth shall be repainted at least once in a three month duration to maintain the black interior, or more frequently as determined by the Government Quality Assurance Representative. If the burning time of an individual sample is less then 65 secs. or the average luminosity of less than 703,000 candlepower, or an individual luminosity is less than 670,000 candlepower, the mix/batch shall be rejected.

4.5.3 Hardness test of canister body. A gooseneck adapter shall be used to permit taking a Rockwell hardness reading on the inside surface of the canister body approximately one inch from the open end. Two hardness determinations, taken 180 degrees apart, shall be taken on each canister body. Each determination shall consist of the average of two readings taken approximately .250 inch apart. The test shall be performed as specified in Method 243.1 of FED-STD-151. After testing, conforming samples may be returned to production.

4.5.4 Proving ground. The projectile assembly shall be assembled with accepted components; fuze, expelling charge, propelling charge, ignition cartridge, rotating disc, pressure plate and obturating mechanism. The cartridge shall be fired from a 4.2" mortar for which the round is standard and in accordance with Table II. However, if meteorological

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conditions or time of fuze functioning so dictate, the proving ground shall adjust the fuze in order to increase the height of burst and insure effective illumination (see 6.). Observation shall be made for compliance with the requirements specified for security of parts, parachute functioning and effective ilimination (see 6.6). In addition, muzzle velocity, time of fuze functioning and height of first light shall be recorded for information only. Any malfunction occuring during the test shall be reported in the firing record. Meteorological data shall also be reported in the official firing record.

TABLE II

Propellant Phase	Fuze Setting* (sec.)	Mortar Elevation* (mils)
12	8.5	900
25 1/2	26.0	800
41	32.0	800

* Fuze setting and elevation is nominal

4.5.5 Salt spray test. This test shall be a 5 percent salt spray test as specified in FED-STD-151, Method 811.1.

5. PACKAGING

5.1 Packing. Packing shall be as specified on dwg. 8848553.

5.2 Marking. Marking shall be as specified on dwg. 8861520.

6. NOTES

6.1 Intended use. The component covered by this specification is intended for use on the 4.2 Inch, Illuminating, M335A2 Cartridge.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, NJ 07801.

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6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARRADCOM.

6.5 Satisfactory functoning. Satisfactory functioning shall be defined as ejection of the parachute and illuminating canister assembly and proper suspension and ignition of the candle.

6.6 Effective illumination. Effective illumination is defined as burning of the candle without exploding or breaking-up while suspended by the parachute. The time recorded for "effective illumination" shall be considered as the time from the appearance of first light from the candle to either (1) burnout of the candle while still in the air (normal burn-out) or (2) burning of the candle until ground impact. Burning of the candle after striking the ground shall not be considered as "effective illumination". Any occurancs which detracts from "effective illumination" as herein defined, shall be recorded on the official firing record.

6.7 Mix/Batch. A mix of candles is defined as that quantity of candles which has been loaded with one mix of illuminant for all increments. A batch is defined as a quantity of candles produced by a mixing process intended to make the product essentially uniform.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-P-12764G Projectile, 4.2 Inch, Illuminating, M335A2 LAP

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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